

MOHAMED CHERIF BRAHAM

linkedin.com/in/mohamedcherif-braham | mohamed-cherif-braham.vercel.app | mohamedcherif.braham@duke.edu | (919) 672-3250

EDUCATION

Duke University, Durham, NC

Expected May 2029

B.S. in Electrical and Computer Engineering and Computer Science | Certificate in Robotics and Automation | GPA: 3.74

Coursework: Engineering Design & Communication, Fundamentals of ECE, Computer Architecture, Data Structures & Algorithms

UWC Adriatic, Duino, Trieste, Italy

Graduated May 2025

SKILLS

Hardware & Electronics: Arduino, Raspberry Pi, ESP32, Seeed Studio boards, Embedded Systems, Digital Logic / RTL Simulation (SystemVerilog, Icarus Verilog), PCB Design (KiCad), Sensor Integration, Bench Debugging (multimeter), Forward/Inverse Kinematics

Mechanical Design & Fabrication: CAD (SolidWorks, Fusion 360, Onshape), tolerance & fit iteration, 3D Printing (FDM), Waterjet/Laser/CNC Cutting, Silicone Molding, BOM Development

Programming: Python, C/C++, Java, JavaScript/TypeScript, Arduino

AI/ML & Computer Vision: PyTorch, TensorFlow, YOLO/YOLOE, Pose Estimation, Edge AI / TinyML

Tools: Git, Linux, ROS2, Docker, Hugging Face, AI-assisted dev (Claude)

Languages: Arabic (Native), French (Fluent), Italian (Proficient), English (Fluent)

RESEARCH & WORK EXPERIENCE

Duke Biomechanics and Wearable Robotics Lab, Duke University

2026 – Present

Research Assistant (Faculty Mentor: Dr. Krithika Swaminathan), Durham, NC

- Developing an OpenExo-based ankle exoskeleton for gait rehabilitation; building the mechanical skeleton and integrating sensing and control.
- Set the exoskeleton's mechanical requirements including motor torque specifications, then reconciled three overlapping vendor BOMs against them into a single purchasing document.
- Selected and validated IMU-based gait-event sensing against published biomechanics literature.
- Leading a team of 4 students on the assembly and wiring process of the exoskeleton.

Innovation Co-Lab, Duke University (OIT)

Feb 2026 – Present

Makerspace Technician, Durham, NC

- Operating and maintaining 30+ 3D printers, waterjets, and laser cutters across Duke's three Innovation Co-Lab fabrication facilities.
- Resolving recurring 3D-printer failures (jammed filament, clogged nozzles, filament changes); teaching 20+ students, faculty, and staff to safely operate the waterjet and laser cutters.

Independent Research – Robotic Manipulation & Kinematics (SO-100 Robotic Arm)

Aug – Sep 2026

Self-directed

- Derived rotation matrices and homogeneous transforms from first principles; implemented a closed-form planar IK solver exploiting the arm's parallel joint geometry. Verified against forward kinematics over 300 simulated round-trip trials with near-zero residual error; simulation-only, not tested on hardware.
- Deployed a zero-shot pick-and-place system on the physical arm: integrated a Claude agent (via MCP) for target estimation with camera calibration and an IK control loop driving arm motion. Achieved repeated success (cube into bucket) with no task-specific training.
- Separately developing an on-arm vision pipeline (YOLOE-seg + ArUco visual servoing) as an alternative to agent-based control; in progress, not yet reliable.

ATAST Organization, Tunisia

Nov 2020 – Aug 2023

AI and Robotics Instructor

- Instructed 15+ students in Python, Arduino, and computer vision (YOLO); supervised engineering-fair projects and 2 middle-school VEX Robotics teams through the build process.

SELECTED TECHNICAL PROJECTS

Blood Loss Monitoring Device for PAS Surgery

- Designed the fluid-collection cup and pumping mechanism in Onshape; silicone-molded the cup and assembled the prototype and harness.
- Validated to within 5 mL absolute error (<0.7% relative error) across a tested range of 250–1000 mL; prototype reviewed and approved by Dr. Sara Jones (Duke Hospital), who advised the team to pursue clinical certification.

ImpaQ – Smart Impact-Sensing Mouthguard

- Built a wearable impact sensor (ESP32, accelerometer) interfaced with a Flutter app for real-time TBI-risk monitoring (C++, Flutter).
- Diagnosed a Bluetooth communication failure on the board: used a multimeter to isolate insufficient power delivery from the onboard voltage regulator, then replaced it with one properly sized for the Bluetooth module, restoring reliable wireless communication.

do_robotics – Phone-as-Robot-Controller Platform

- Designed a custom chassis in Fusion 360 around the phone, motor, and battery geometry; 3D-printed and assembled the build.
- Achieved real-time visual tracking and following of a physical target (a bottle) on a minimal platform: ESP32, battery, and 2 motors only, no onboard compute.
- Ran all vision/ML on-device via edge ML (TFLite) on a smartphone, sending commands to the ESP32 over Bluetooth/USB-serial via a custom 5-byte protocol; built a block-based visual scripting interface in the companion app.

Cache-Miss Tracker FSM – Digital Design in SystemVerilog

Optional coursework assignment

- Designed a 6-state Moore FSM tracking outstanding cache misses, then extended it to a multi-entry tracker array with per-entry tag comparators; verified both with self-checking testbenches in Icarus Verilog (13/13 and 14/14 checks passing).

Earlier Independent CV/Robotics Projects – WildGuard & Oil-Detecting UUV

- WildGuard (Wildfire Detection, Oct 2021 – Apr 2022):** Built a real-time CV/ML pipeline (YOLO on Raspberry Pi) fused with environmental sensor data and a Python/Pandas time-series pipeline; ~83% detection accuracy across varying lighting and distances.
- Oil-Detecting UUV (Autonomous Spill Response, Jun 2023):** Built a computer-vision pipeline (YOLO, Raspberry Pi) for real-time oil-spill detection from the vehicle's onboard camera, paired with closed-loop steering control to autonomously navigate toward spill locations.

AWARDS & LEADERSHIP

- Duke University International Scholar (2025–2029)
- STEM awards: Nat'l Rep., Expo Science Asia Dubai (2022); Gold Medal & Top-10, I-FEST² (2023); Bronze, Genius Olympiad NY (2023); Gold, I Giovani e le Scienze (2024); APA Award (2024).